

MARS

Malware Analysis & Reverse-engineering System

Analysis ID: RL_SMOKE_001

Target: malware_sample.exe

Generated: 2026-06-19 12:00:00

Table of Contents

Table of Contents	2
1. File Intake Summary	3
2. Package & Archive Unpacking	3
Extracted Artefacts	3
3. Deep Static Analysis	3
Target: malware_sample.exe	3
4. Dynamic Sandbox Analysis	4
Target: malware_sample.exe	4

1. File Intake Summary

Raw metadata captured by the intake module when the sample was submitted. Hash values are computed immediately on ingestion before any analysis modifies memory.

Analysis ID	RL_SMOKE_001
Filename	malware_sample.exe
File Path	/tmp/malware_sample.exe
Extension	.exe
File Size	40960 bytes
MD5	aabbccdd11223344
SHA1	aabbccdde1122334455
SHA256	aabbccdde112233445566778899aabbccdde112233445566778899aabb1122
Submission Timestamp	2026-06-19 12:00:00

2. Package & Archive Unpacking

Total Extracted Files	2
Flagged Payloads	1

Extracted Artefacts

Files unpacked from the archive. FLAGGED entries have extensions that match executable payload types and are forwarded to the static analysis phase.

Path	Ext	Size (B)	SHA-256 (abbrev.)	Flag
dropper/payload.exe	.exe	8192	deadbeefdeadbeefdead...	FLAGGED
config/settings.ini	.ini	512	cafebabecafebabecafe...	—

3. Deep Static Analysis

PE file structure analysis performed without executing the sample. Results include header metadata, security mitigations, section entropy, IAT inspection, manifest parsing, string extraction, and YARA signature matching.

Target: malware_sample.exe

PE Headers

Machine Architecture	0x14c
Compile Timestamp	2025-01-15 08:30:00
DOS Header Offset (e_lfanew)	0xe8
Address of Entry Point	0x1234
Image Base	0x400000
Number of Sections	5

Mitigations

DEP / NX Bit	Enabled
ASLR	Disabled
CFG (Control Flow Guard)	Disabled
Stack Canaries (/GS)	Disabled

Sections

Section: .text	Perms: R-E Entropy: 6.12
Section: .data	Perms: RW- Entropy: 7.85 [PACKED/HIGH ENTROPY]
Section: .reloc	Perms: R-- Entropy: 3.20

Suspicious Imports

Tracked APIs Found	3
APIs	CreateRemoteThread, VirtualAllocEx, WriteProcessMemory

Manifest Data

Manifest Status	Parsed Successfully
Requested Execution Level	requireAdministrator

Strings Analytics

IPv4	3
IPv6	0
URL	2
Registry	5
Email	0
Password-Like	1

YARA Signatures

Hits	2
Matched Rules	Mitre_T1055_ProcessInjection, Ransomware_Generic

4. Dynamic Sandbox Analysis

Behavioural telemetry captured during live detonation inside the VMware sandbox. The agent instruments filesystem, registry, network, process, memory, and hardware subsystems in real time. Events are classified and counted per category.

Target: malware_sample.exe

Telemetry Summary

Category	Events	Notable / High-Risk
Filesystem Activity (FR-DYN-01)	2	1 flagged
Registry Mutations (FR-DYN-02)	1	—

Category	Events	Notable / High-Risk
Persistence Mechanisms (FR-DYN-03)	1	1 flagged
Process Activity (FR-DYN-04)	2	1 flagged
Memory Forensics (FR-DYN-05)	0	—
Network Telemetry (FR-DYN-06)	3	3 flagged
Hardware / System Stress (FR-DYN-07)	0	—
System & Agent Events	2	—

Event Detail

Filesystem Activity (FR-DYN-01) (2 events)

- **FILE_DROP: C:/Windows/Temp/svchost32.exe**
- FILE_WRITE: C:/Users/Admin/AppData/Roaming/mal.dat

Registry Mutations (FR-DYN-02) (1 events)

- REG_SET: HKCU/Software/Microsoft/Windows/CurrentVersion/Run/updater

Persistence Mechanisms (FR-DYN-03) (1 events)

- **REG_RUN_KEY: HKCU Run -> C:/Windows/Temp/svchost32.exe**

Process Activity (FR-DYN-04) (2 events)

- **PROCESS_SPAWN: cmd.exe /c whoami**
- **PROCESS_INJECT: lsass.exe (PID 840)**

Network Telemetry (FR-DYN-06) (3 events)

- **DNS: evil-c2.example.com**
- **HTTP: GET http://evil-c2.example.com/payload.bin**
- **TLS: 192.168.1.99:443**

System & Agent Events (2 events)

- Agent started
- UAC bypass attempted